

## Experiment No: 6

**Title:**

To study Agile Methodology and Test Case Management using JIRA Tool.

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**Aim:**

To understand Agile methodology and manage project tasks and test cases using the JIRA tool.

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**Theory:**

Agile methodology is an iterative software development approach that focuses on collaboration, flexibility, customer feedback, and incremental delivery of working software. It divides the project into smaller modules called sprints, allowing teams to continuously improve product quality and respond quickly to requirement changes.

JIRA is a project management and issue-tracking tool widely used in Agile environments. It helps teams plan sprints, create user stories, assign tasks, track bugs, manage workflows, and monitor project progress efficiently. JIRA supports Scrum and Kanban boards that allow visualization of task status such as To Do, In Progress, and Completed.

Test case management in JIRA allows tracking of testing activities by organizing issues as tasks, bugs, or stories and monitoring their lifecycle through workflow stages. This ensures better quality assurance and defect tracking throughout software development.

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**Requirements:**

- JIRA Software account
  - Internet connection
  - Web browser
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**Procedure:**

**Step 1:** Login to JIRA and create a new project workspace.

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**Step 2:** Create project tasks such as Requirement Analysis, System Research, Design, Development, Testing, and Deployment.

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**Step 3:** Organize tasks into Agile workflow stages like To Do, In Progress, and Complete using board view.

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**Step 4:** Assign tasks to team members and set priorities for better task tracking.

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**Step 5:** Monitor project progress using the Agile board tracking system in JIRA.

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**Step 6:** Manage testing-related activities by tracking issues through workflow stages.

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**Result:**

The image displays two screenshots of the Jira Gmail board interface, showing the progression of tasks over time.

**Top Screenshot:**

- Spaces / Gmail:** The board is titled "GMAIL board".
- Navigation:** Includes Summary, Timeline, Backlog, Active sprints, Calendar, Reports, List, Forms, Goals, Components, Development, Code, Releases, and More.
- Filters:** Search board, Epic, Type, Quick filters.
- Columns:** TO DO, IN PROGRESS, DONE.
- Tasks:**
  - TO DO:** Verify login functionality (VERIFY GMAIL FUNCTIONALITY), GMAIL-5.
  - IN PROGRESS:** (Empty)
  - DONE:** Analyse requirement (GMAIL-7), Design text cases (GMAIL-8), Execute text cases (GMAIL-9).

**Bottom Screenshot:**

- Spaces / Gmail:** The board is titled "GMAIL board".
- Navigation:** Includes Summary, Timeline, Backlog, Active sprints, Calendar, Reports, List, Forms, Goals, Components, Development, Code, Releases, and More.
- Filters:** Search board, Epic, Type, Quick filters.
- Columns:** TO DO, IN PROGRESS, DONE.
- Tasks:**
  - TO DO:** (Empty)
  - IN PROGRESS:** Verify login functionality (VERIFY GMAIL FUNCTIONALITY), GMAIL-6; Verify inbox functionality (VERIFY GMAIL FUNCTIONALITY), GMAIL-4.
  - DONE:** (Empty)

The image displays two screenshots of the Jira web application interface, specifically the 'GMAIL board' within the 'Gmail' space.

**Top Screenshot (Backlog View):**

- Header:** Shows 'Spaces / Gmail', 'GMAIL board', and navigation tabs: Summary, Timeline, Backlog (active), Active sprints, Calendar, Reports, List, Forms, Goals, Components, Development, Code, Releases, and More.
- Left Sidebar:** Contains navigation links like 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', 'Starred', 'Recent', 'Gmail', 'GMAIL board', 'My Software Team', 'More spaces', 'Recommended', 'Collect requests', 'Filters', 'Dashboards', 'Goals', 'Teams', and 'More'.
- Main Content:**
  - Search backlog:** Includes a search bar, version filter (Epic), quick filters, and clear filters.
  - Epic List:** Shows 'No epic' and 'Verify Gmail functionality'.
  - Backlog (5 work items):** A list of tasks under the 'Verify Gmail functionality' epic:
    - GMAIL-3: Verify compose mail functionality (Status: TO DO, Priority: 5)
    - GMAIL-6: Verify login functionality (Status: TO DO, Priority: 3)
    - GMAIL-2: Verify Gmail functionality (Status: TO DO, Priority: 5)
    - GMAIL-4: Verify inbox functionality (Status: TO DO, Priority: 5)
    - GMAIL-5: Verify logout functionality (Status: TO DO, Priority: 5)

**Bottom Screenshot (Sprint View):**

- Header:** Similar to the top screenshot, but the 'Backlog' tab is not active.
- Main Content:**
  - Epic List:** Same as the top screenshot.
  - GMAIL Sprint 2 (24 Feb - 10 Mar (3 work items)):** A list of tasks currently in the sprint:
    - GMAIL-6: Verify login functionality (Status: TO DO, Priority: 3)
    - GMAIL-4: Verify inbox functionality (Status: TO DO, Priority: 5)
    - GMAIL-3: Verify compose mail functionality (Status: TO DO, Priority: 5)
  - Backlog (2 work items):** A list of tasks not in the current sprint:
    - GMAIL-2: Verify Gmail functionality (Status: TO DO, Priority: 5)
    - GMAIL-5: Verify logout functionality (Status: TO DO, Priority: 5)

Successfully studied Agile methodology and performed task tracking and workflow management using the JIRA tool.

**Conclusion:**

Thus, Agile methodology and test case management were successfully implemented using the JIRA tool by creating tasks, organizing workflow stages, assigning priorities, and monitoring project progress efficiently.